Table 1: Injection-lens candidates for the squeezing cavity (input waist $w_0 = 1.35$ mm at $z = 0$, ROC = 100 mm, $\theta = 13.22°$). Positions z are measured from the input waist; M1 sits at $z = 1635$ mm. The overlap is the total (tangential × sagittal) power overlap with the cavity eigenmode; the maximum reachable with spherical lenses is 0.9230. The last two columns give the worst overlap when any single lens is displaced by ±1 mm and ±5 mm.

Lenses: f (mm) @ z (mm)	Overlap	±1 mm	±5 mm	Verdict
Two lenses				
35 @ 218.5, 50 @ 305.2	0.9230	0.027	0.000	unusable (too sensitive)
50 @ 953.6, 35 @ 1040.3	0.9230	0.180	0.001	unusable (too sensitive)
1000 @ 729.3, 750 @ 1577.1	0.8528	0.852	0.847	low overlap, lens at limit
400 @ 60.0, 300 @ 904.6	0.8122	0.810	0.797	low overlap
500 @ 60.0, 300 @ 1027.1	0.8007	0.798	0.787	low overlap
Three lenses				
500 @ 139.3, 300 @ 852.4, 400 @ 1174.0	0.9230	0.923	0.917	good
500 @ 405.8, 400 @ 1124.2, 300 @ 1241.5	0.9230	0.923	0.918	good
1000 @ 107.5, 750 @ 1406.2, 300 @ 1417.3	0.9230	0.923	0.922	good (two lenses 11 mm apart)
400 @ 67.2, 50 @ 498.4, 300 @ 978.6	0.9230	0.922	0.902	good

Table 2: Lens set written into the main script (`bowtie_mode_matching.py`): input waist $w_0 = 1.35$ mm, ROC = 100 mm, positions measured from the input waist (M1 at $z = 1635$ mm).

Lens	f (mm)	Position z (mm)	Beam radius on lens (mm)
L1	400	400.4	1.35
L2	300	1053.0	0.86
L3	500	1108.9	0.89

Distance L1–L2: 652.6 mm; L2–L3: 55.9 mm; L3 to M1: 526.1 mm
Overlap, single-plane model of the script: 1.0000
Overlap with the 13.22° tilt (astigmatic): 0.9211 (ceiling 0.9230)
Worst overlap for a ±5 mm displacement of any one lens: 0.993